a result of the first transfer, NADH is oxidized to NAD+, and FMN is reduced to $FMNH_2$.

In the second step, $FMNH_2$ passes 2 H+ to the other side of the mitochondrial membrane and passes 2 $e^-$ to coenzyme Q. As a result, $FMNH_2$ is oxidized to FMN. Coenzyme Q also picks up additional 2H+ from the surrounding aqueous and releases to other side of the membrane.

In the next step, electrons are passed successively from coenzyme Q to cyt $b_1$, cyt $c_1$, cyt c, cyt a, cyt $a_3$. Each cytochrome in the chain is reduced, as it picks up electrons and is oxidized as it gives up electrons. The last cytochrome cyt a3 passes its electrons to molecular $O_2$ which picks up protons from the surrounding medium to form $H_2O$.

$FADH_2$ derived from the Krebs cycle is another source of electrons. Thus at the end of ETC, NADH pumps three protons (synthesizes 3ATPs) whereas $FADH_2$ pumps only two protons (synthesizes 2ATPs).

### 4.6.1 Chemiosmotic Mechanism of ATP

Chemiosmotic mechanism of ATP synthesis was first proposed by the Biochemist, Peter Mitchell in 1961. In ETC, when energetic electrons from NADH pass down the carriers, some of the carriers (proton pumps) in the chain pump [actively transport] protons across the membrane to inner membrane space. Thus in addition to a concentration gradient, an electrical charge gradient is created. The resulting electro chemical gradient has potential energy called proton motive force.

The proton diffuses across the membrane through protein channels that contain an enzyme called ATP synthase. When this flow occurs, energy is released and is used by the enzyme to synthesize ATP from ADP and phosphate.

At the end of the chain, electrons join with protons and $O_2$ in the matrix fluid to form $H_2O$. Thus $O_2$ is the final electron acceptor. ETC also operates in photophosphorylation and is located in thylakoid membrane of Cyanobacteria (BGA), and of eukaryotic chloroplasts.

Overview of Aerobic respiration (Figure 4.7):

- Electron transport chain regenerates NAD and FAD which can be used again in Glycolysis and Krebs cycle.
- Various electrons transfer in the electron transport chain generates about 34 ATP, (10 NADH = $10 \times 3 = 30$ + 2 $FADH_2 = 2 \times 2 = 4$).
- A total of 38 ATP molecules is generated from one molecule of glucose oxidized in prokaryotes, whereas in eukaryotes, 36 molecules of ATP is generated because in eukaryotes, some energy is lost when electrons are shuttled across the mitochondrial membranes that separate Glycolysis (in the cytoplasm) from the electron transport chain (Table 4.2). There is no such separation exists in prokaryotes.

$$C_6H_{12}O_6 + 6CO_2 + 38ADP + 38P_i \longrightarrow 6CO_2 + 6H_2O + 38\ ATP$$



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